
Methods to Obtain Experimental Data

1. Determination of the Time Period of Small Oscillations About ϕ_f

1.1. Aim

To measure the time period T of small oscillations about the equilibrium angular displacement ϕ_f in the presence of the external magnetic field.

1.2. Apparatus

- Stand
- Cylindrical NdFeB magnet (diameter = 20 mm, length = 20 mm)
- Cotton twine
 - Length of string between the two rings: $\frac{1}{2}$ inch
 - Length of string joining the outer ring to the support: 20 cm
- Mild steel ring (diameter = 3 inches, thickness = 3mm)
- Mild steel ring (diameter = 4 inches, thickness = 3mm)

1.3. Procedure

1. Estimate the natural angular displacement ϕ_0 in the absence of the external magnetic field, where the torsional torque is zero, by hanging the tied rings under gravity.
2. Arrange the experimental setup by attaching the magnet to the stand and suspending the ferromagnetic rings.
3. Estimate the natural angular displacement ϕ_f in the presence of the magnetic field, where the net torque is zero.
4. Keeping the outer ring fixed at the equilibrium position ϕ_f , displace the inner ring by a small angle about ϕ_f .
5. Measure the time for 5 oscillations of the inner ring about ϕ_f .
6. Divide the total time by 5 to obtain the time period T of one oscillation.
7. Repeat the measurements multiple times (10 trials) and calculate the average time period T .

1.4. Observations

In our setup, ϕ_0 was found to be $\frac{\pi}{2}$, leading to $\phi_f = \frac{\pi}{2}$ as predicted theoretically.

2. Determination of Time Period T for Small Oscillations for thicker rings

2.1. Aim

To measure the time period T of small oscillations about the equilibrium angular displacement ϕ_f in the presence of the external magnetic field, for thicker rings.

2.2. Apparatus

- The same apparatus as in the previous experiment, with the following change:
 - Thickness of the rings: 5 mm each.

2.3. Procedure

1. Estimate the natural angular difference ϕ_0 in the absence of the magnetic field, where the torsional torque is zero, by hanging the tied rings under gravity.
2. Arrange the experimental setup by attaching the magnet to the stand and suspending the ferromagnetic rings.
3. Estimate the natural angular difference ϕ_f in the presence of the magnetic field, where the net torque is zero.
4. Keeping the outer ring fixed at the equilibrium position ϕ_f , displace the inner ring by a small angle about ϕ_f .
5. Measure the time for 5 complete oscillations of the inner ring about ϕ_f .
6. Divide the total time by 5 to obtain the time period T of one oscillation.
7. Take the average of multiple readings (10 trials were conducted in this experiment).

2.4. Observation

In our setup, we observed that $\phi_0 = \frac{\pi}{2}$, which causes $\phi_f = \frac{\pi}{2}$ as well. This agrees with our theoretical predictions.

3. Determination of Time Period T for Large Oscillations as a Function of $\Delta\phi$

3.1. Aim

Find the time period T for large oscillations as a function of $\Delta\phi$.

3.2. Apparatus

- Stand
- Cylindrical NdFeB magnet (diameter = 20 mm, height = 20 mm)
- Cotton twine
 - Length of string between the two rings: $\frac{1}{2}$ inch
 - Length of string joining the outer ring to the support: 20 cm
- Mild steel ring (diameter = 3 inches)
- Mild steel ring (diameter = 4 inches)

3.3. Procedure

1. Estimate the natural angular difference ϕ_0 in the absence of the magnetic field, where the torsional torque is zero, by hanging the tied rings under gravity.
2. Arrange the experimental setup by attaching the magnet to the stand and suspending the ferromagnetic rings.
3. Estimate the natural angular difference ϕ_f in the presence of the magnetic field, where the net torque is zero.
4. Keeping the outer ring fixed at the equilibrium position ϕ_f , displace the inner ring by a certain angle $\Delta\phi$ about ϕ_f .
5. Measure the time for 5 complete oscillations of the inner ring about ϕ_f .
6. Divide the total time by 5 to obtain the time period T of one oscillation for the given $\Delta\phi$.
7. Repeat the procedure for other values of $\Delta\phi$, gradually increasing it.
8. Take the average of multiple readings (10 trials were conducted in this experiment).

4. Determination of ϕ_f as a Function of ϕ_0

4.1. Aim

To obtain the relation of ϕ_f as a function of ϕ_0 .

4.2. Apparatus

The same apparatus as in the first experiment, using rings with a thickness of 3 mm.

4.3. Procedure

1. Tie the string such that the natural angular difference ϕ_0 in the absence of the magnetic field, where the torsional torque is zero, is varied. This can be done in intervals of 15 degrees.
2. A possible method to achieve this:
 - Give the string a slight twist and use an adhesive such as Fevikwik while both rings are aligned.
 - After taking each reading, scrape off the adhesive to adjust the angle for the next measurement.
 - This method is easier to control compared to tying, as tying often results in $\phi_0 = \frac{\pi}{2}$.
3. For each reading:
 - Estimate the natural angular difference ϕ_0 in the absence of the magnetic field, where the torsional torque is zero, by hanging the tied rings under gravity.
 - Arrange the experimental setup by attaching the magnet to the stand and suspending the ferromagnetic rings.
 - Estimate the natural angular difference ϕ_f in the presence of the magnetic field, where the net torque is zero. This can be observed by noting the angle at which the rings come to rest after giving the inner ring a small displacement.

5. Study of Angular Evolution Using Motion Tracking and Video Analysis

5.1. Aim

To study the evolution of θ_1 , θ_2 , and ϕ with time using a motion tracker and video analysis software.

5.2. Apparatus

- Same apparatus as in the first experiment.
- Tracker Video Analysis and Modeling Tool for Physics Education¹.

5.3. Procedure

1. Position a camera vertically above (or below) the rings and take a video in slow motion.
2. Apply a large impulse to either of the rings and record the top (or bottom) view of the dynamics of the rings.

5.4. Notes

- Add clearly visible markers to the rings (we used black bindis).
- For easy tracking, use a clear background.
 - In our setup, we recorded the video from below but were unable to achieve a perfectly vertical alignment. As a result, the path of the bindi appeared elliptical rather than circular.
 - We placed white paper between the magnet and the topmost point of the upper ring to enhance visibility.

¹<https://physlets.org/tracker/>